

TOPIC BREAKDOWN

Project Title: A Predictive Model for Student Dropout Risk Using Machine Learning

Background / Problem Statement

Student dropout is a major challenge in higher institutions. Traditional methods of identifying at-risk students rely on manual monitoring and may fail to detect early warning signs. This project aims to use machine learning techniques to analyse student data and predict dropout risk, enabling early intervention and better decision-making.

Aim: To design and implement a machine learning-based system for predicting student dropout risk at Umaru Musa Yar'adua University, Katsina.

Specific Objectives

- Collect and preprocess student academic and demographic data.
- Develop and train machine learning models for dropout risk prediction.
- Evaluate model performance using accuracy, precision, recall, and F1-score.
- Develop a web-based system for viewing student risk predictions and reports.

Scope & Limitations

Scope: The system will analyse factors such as CGPA, attendance, carryovers, fees payment status, student level, age, and other student information to classify dropout risk and generate reports.

Limitations: The system will not include psychological factors, real-time student behaviour monitoring, or integration with existing university management systems.

Methodology / Approach

The Agile Software Development Methodology and Machine Learning workflow will be adopted:

- **Research:** Review existing dropout prediction systems and collect requirements.
- **Data Preparation:** Clean, transform, and prepare student datasets.
- **Model Development:** Train and evaluate machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest.

Implementation & Testing: Develop the web system and test its functionality and performance.

Expected Outcome / Deliverables

- A trained machine learning model for student dropout prediction.
- A functional web-based dropout risk prediction system.
- Student risk dashboard with reports and analysis.
- Project documentation and final system demonstration.

Tools & Materials Needed

Software: Python, Scikit-learn, Pandas, Flask, React.js, Node.js, Express.js, MongoDB, Visual Studio Code, Git/GitHub, Postman.

Hardware: Laptop computer and internet connection.

Estimated Budget: ₦50,000 - ₦100,000 for development, documentation, hosting, and related expenses.

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